Gage William Boyd

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EDUCATION

Northern Arizona University — **Expected May 2028**

Bachelors of Science in Mathematics & Bachelors of Science in Computer Science · GPA: 3.6 — *Flagstaff, AZ*

- **Relevant Coursework:** Regression Analysis, Probability & Statistics, Differential Equations, Calculus I-III, Discrete Mathematics, Foundation of Mathematics
- **Leadership & Activities:** Lumberjack Scholarship Recipient · SCLA Honors Society · Association for Computing Machinery · FAMUS (undergraduate mathematics research)
- **Jump Trading Probability Cup**: Top 150 of 3,539 competitors (top ~4%): iteratively refit probabilistic forecasting models in Python, applying de-vigging and extremizing to capitalize on identified edge

EXPERIENCE

Machine Learning Research Assistant — **May 2026 – Fall 2026**

NAU UQLID Lab — Scientific Machine Learning & Uncertainty Quantification | here — *Flagstaff, AZ*

- Built PyTorch neural-network surrogates predicting natural frequencies of structural materials under uncertainty, fusing sparse high-fidelity finite-element data with low-fidelity simulations to approximate full-model accuracy at a fraction of the compute cost
- Trained a high-fidelity model to 97.4% accuracy in predicting bandgap locations, applying backpropagation, gradient descent, and uncertainty quantification to a dataset developed at the Indian Institute of Technology
- Integrated physics-informed neural networks (PINNs) as an additional training layer, embedding governing physical equations to improve predictive accuracy on sparse data

Project Engineering Intern — **Summer 2026**

Achen-Gardner Construction — *Phoenix, AZ*

- Tracked quantities and drafted submittals, purchase orders, subcontracts, RFIs, and LOIs for a $16M public-works intersection project, coordinating sequencing across multiple agencies and municipalities over a 3-month period
- Cross-referenced vendor submittals against pre-approved materials lists, catching 10+ specification discrepancies before procurement
- Drafted Traffic Control plans on Bluebeam and AutoCAD; drafted Guaranteed Maximum Price (GMP) submittals for the City of Buckeye, quantifying work done for a project 7-months from completion

PROJECTS

Monte Carlo Options Pricer & Backtester | *Python, NumPy, pandas, yfinance* · GitHub — **2026**

- Priced European call options via **Monte Carlo** simulation of geometric Brownian motion across 10,000 paths, validating output against the Black-Scholes closed-form solution
- Estimated annualized realized volatility from AAPL log returns, benchmarked simulated prices against live market quotes, and visualized the implied-volatility smile across strikes
- Backtested a 21-month mean-reversion volatility strategy (entering when implied volatility exceeded 1.35× realized) and documented a short-gamma drawdown during the April 2025 tariff shock as a regime-risk finding

Energy Commodities Backtester | *Python, pandas, EIA & FRED APIs* — **2026**

- Built a signal-research framework for WTI crude and natural gas, ingesting EIA and FRED data and implementing momentum and mean-reversion strategies with regime-switching logic
- Evaluated risk-adjusted performance across volatility regimes (Sharpe ratio, max drawdown, hit rate) against a buy-and-hold benchmark

Full-Stack Retirement Monte Carlo Simulator | *React, FastAPI, Python, NumPy* · Live Site — **2026**

- Engineered a 2,000-path GBM simulation engine returning median retirement timeline, success rate, and top/bottom-decile outcomes, with full trajectory storage for frontend visualization
- Deployed a React/Vite frontend on Vercel connected to a FastAPI backend on Render, serving simulation statistics through a REST API

Rotating Ultrasonic Sensor Radar | Arduino, C++, Processing 4 — **2026**

- Programmed embedded C++ firmware and a real-time GUI for a 2D radar sweeping 0–160° at 1° resolution, displaying object position, angle, and distance within a 40 cm range

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, MATLAB, React, SQL, TypeScript

Frameworks & Tools: PyTorch, NumPy, pandas, React, FastAPI, Git, Linux, Vercel, Render, Excel

Interests: Chess, Poker, Literature, Philosophy, Public Policy, Bodybuilding, Brazilian Jujitsu